

Myocarditis, Pericarditis, and Cardiac Tumors

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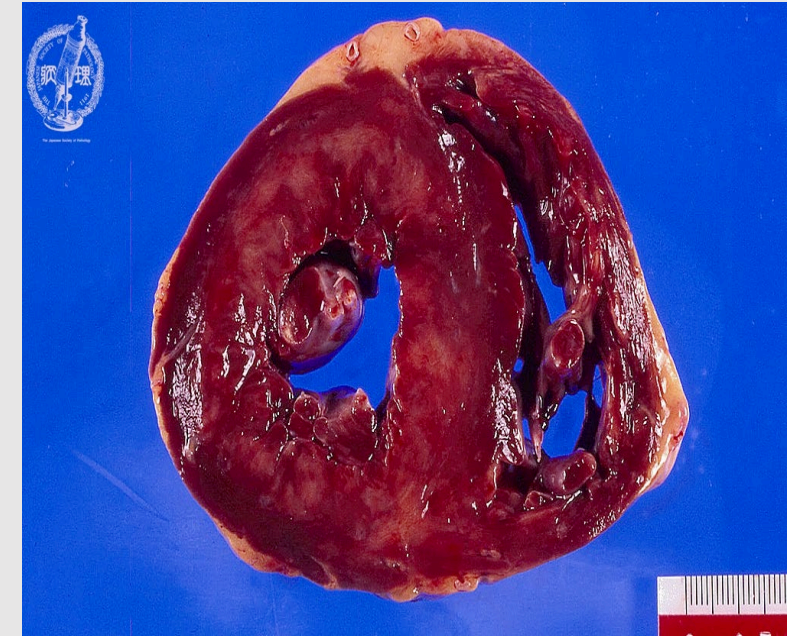


Session Objectives

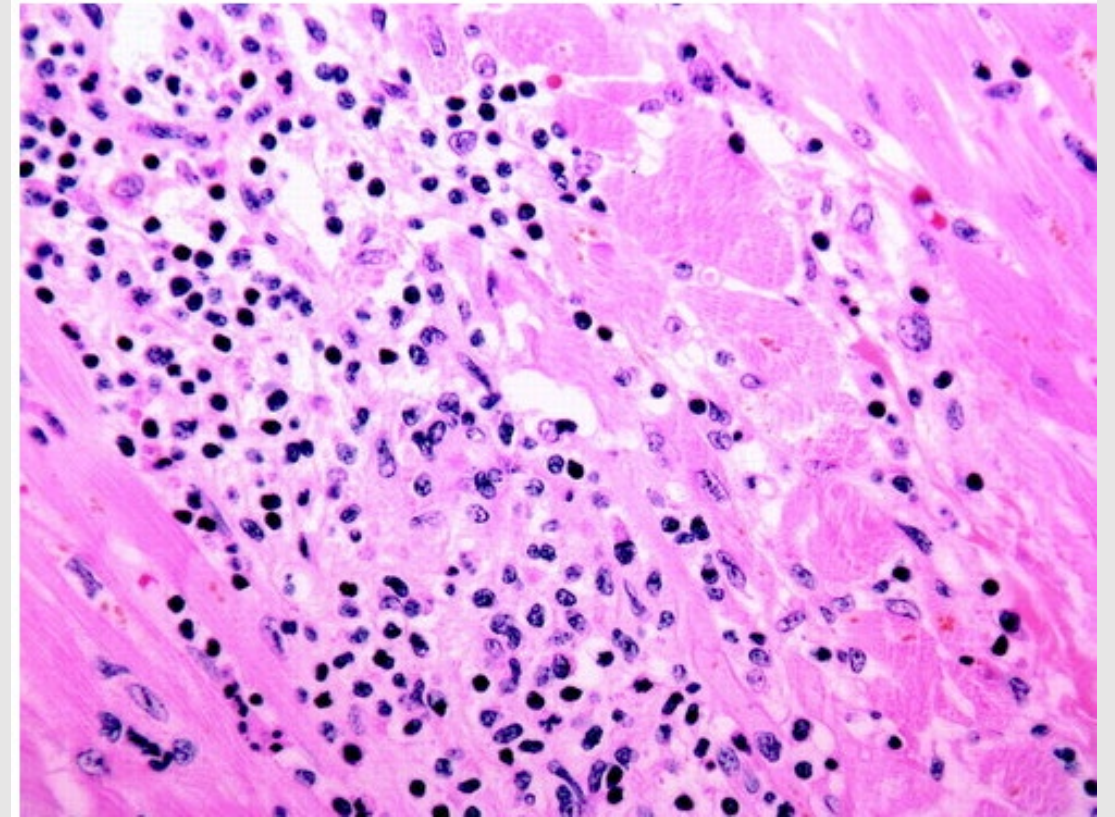
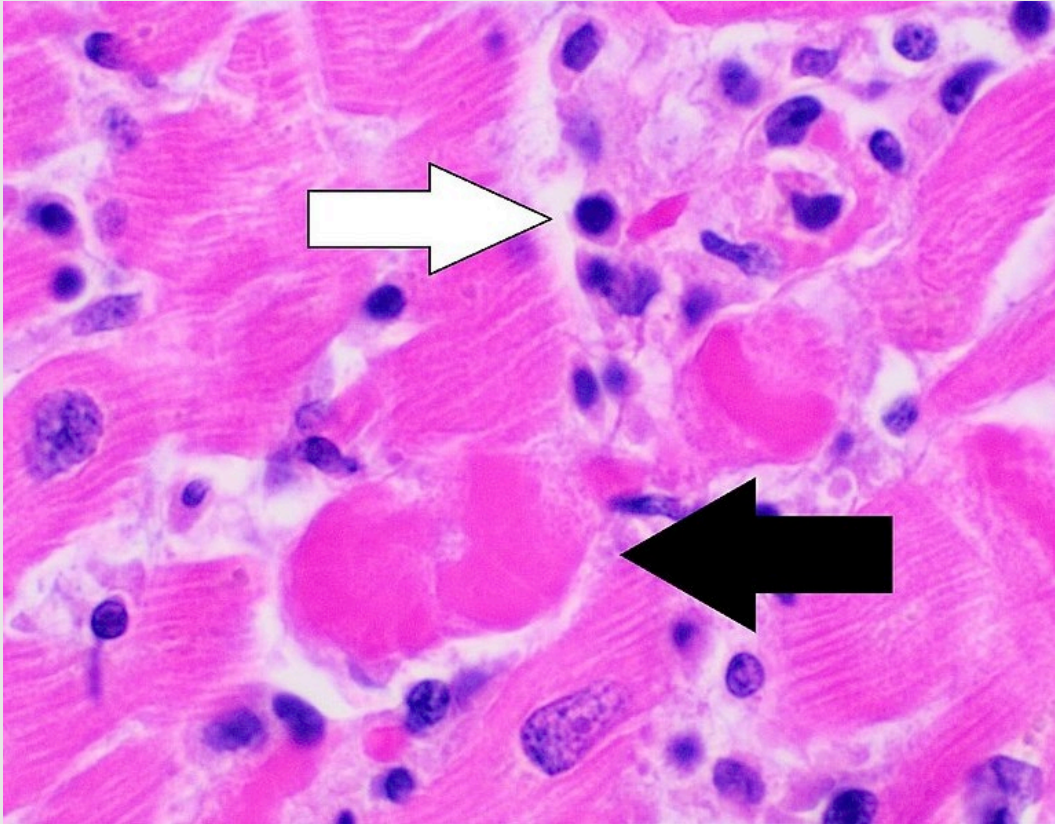
- At the conclusion of the session, students should be able to:
 1. Discuss myocarditis – how it manifests clinically, its causes, and consequences
 2. Grasp the concepts of pericardial diseases and know what causes them as well as describe their morphology and histology
 3. Summarize the various cardiac tumors

Myocarditis

- Definition: **Inflammation of the heart muscle with myocyte necrosis**. It must not be caused by ischemic heart disease (ex. MI) to be classified as myocarditis. (Dallas Criteria)
- Clinical presentation can range from **mild/self limited to fulminant** with severe heart failure
- 50% of cases are idiopathic
- When an etiology is identified, it is most commonly caused by **viruses** but may be caused by bacteria, fungi, parasites, cardiotoxins, drugs, and systemic autoimmune disorders



Histopathology Example





Epidemiology of Myocarditis

- Prevalence is estimated at about 10.2 to 105.6 per 100,000 and annual occurrence of around 1.8 million cases
- Death rate is highest in first year of life
- After age 15 death rate is higher in males than females
- Disability largely due to heart failure
- Can be difficult to diagnose in areas that rely on clinical presentation



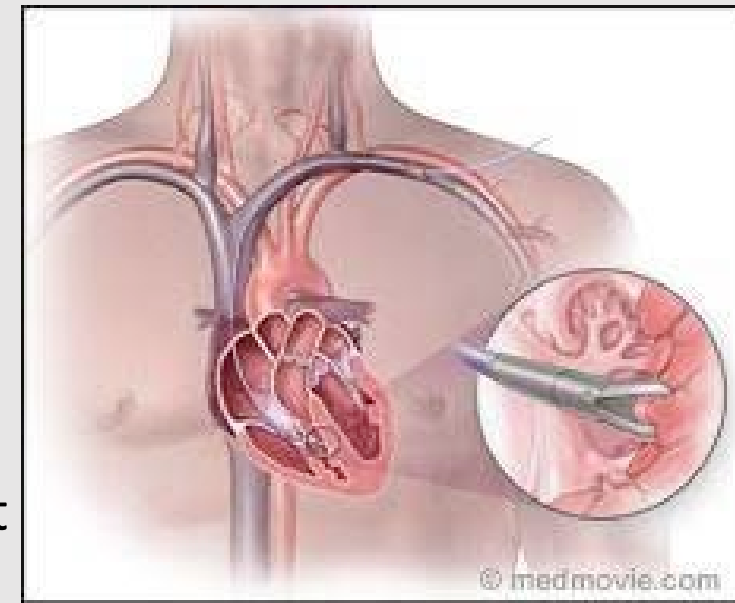
Clinical Myocarditis

- Presentation ranges from asymptomatic with full recovery to heart failure or arrhythmias that may be followed by sudden death
 - Fatigue, dyspnea, palpitations, precordial discomfort, and fever exist between these two extremes
- Symptoms **may mimic acute MI**
- **Dilated cardiomyopathy** may result as a late complication



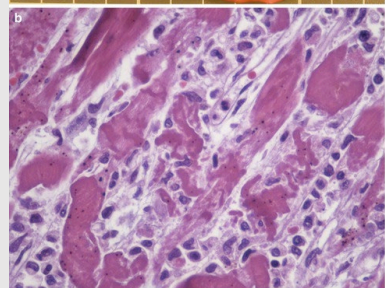
Diagnosis of Myocarditis

- Combination of clues from H & P and diagnostic tests
 - **Endomyocardial Biopsy is gold standard** but only indicated if results significantly impact patient management
 - **Always get:** ECG, CRP, troponins, chest radiograph, Echocardiogram
 - **May need:** coronary angiography if symptomology consistent with ACS or high risk of ischemic heart disease
 - Acute coronary syndrome (CRP/troponin. ↓), Myocarditis (CRP/troponin ↑)
- Cannot make definitive diagnosis of myocarditis without EMB that meets Dallas criteria and is indicated by immunohistochemical stains
- Termed clinically suspected myocarditis if clinical presentation and non-invasive testing is consistent



Clinical Classifications of Myocarditis

- Acute
 - **2-4 weeks after exposure** to etiologic agent (usually a virus)
 - Long term **prognosis is worse than in fulminant myocarditis** due to increased risk of dilated cardiomyopathy
 - Physical exam: 3rd heart sound, mitral/tricuspid insufficiency murmurs, pericardial friction rub if concomitant pericarditis, signs of heart failure
- Fulminant
 - Occurs usually **within 2 weeks after exposure** to etiologic agent
 - **Profound left ventricular dysfunction**
 - Multiple foci of active inflammation and necrosis
 - **Either die within 2 weeks or fully recover**



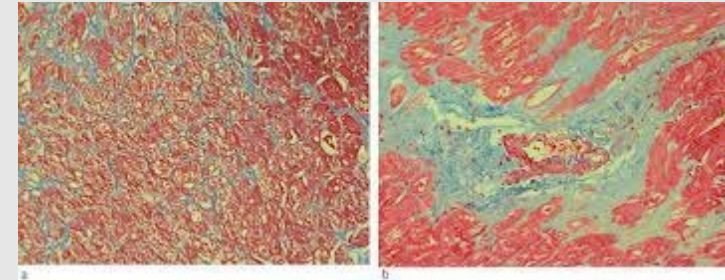
Clinical Classifications Cont.

- **Chronic Active**

- Indistinct onset
- Symptoms >2 weeks
- **Moderate ventricular dysfunction** on presentation and active or borderline myocarditis on biopsy
- Display **ongoing inflammation and fibrosis; develops into restrictive cardiomyopathy** 2 to 4 years after presentation

- **Chronic Persistent**

- Persistent histologic infiltrate, often with foci of myocyte necrosis
- **No ventricular dysfunction** despite other cardiovascular symptoms (chest pain/palpitations)





Causes of Myocarditis

- Viral
- Bacterial
- Fungal
- Parasitic
- Immune Mediated
- Direct cardiotoxins

Viral Myocarditis

- **Most common cause of myocarditis** when an etiology is identified
- Mostly caused by **Enteroviruses notably Coxsackie B Virus**
 - Parvovirus B-19 and HHV-6 are also major viral causes
 - Less common: CMV, HIV, and influenza
- Causative virus can be identified via serologic studies or rarely viral nucleic acid sequences in myocardial biopsies
- Damage to myocardium by:
 - Direct cytopathic effects of the virus
 - Immune response to virus



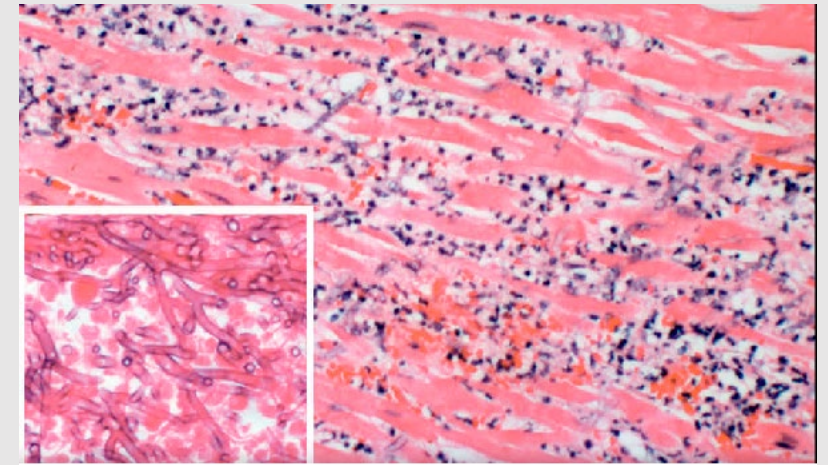
Bacterial Myocarditis

- Bacterial Infections that can cause myocarditis
 - Chlamydiae
 - Rickettsiae
 - Neisseria meningitidis
 - **Borrelia burgdorferi (Lyme carditis)**
 - Spirochetes infiltrate the heart tissue causing an inflammatory response
 - AV block is classic presentation
 - **Corynebacterium diphtheriae**
 - Diphtheria Toxin (subunit B binds to cells allowing subunit A to be released in cytosol and inhibit EF2 which is important for protein synthesis)



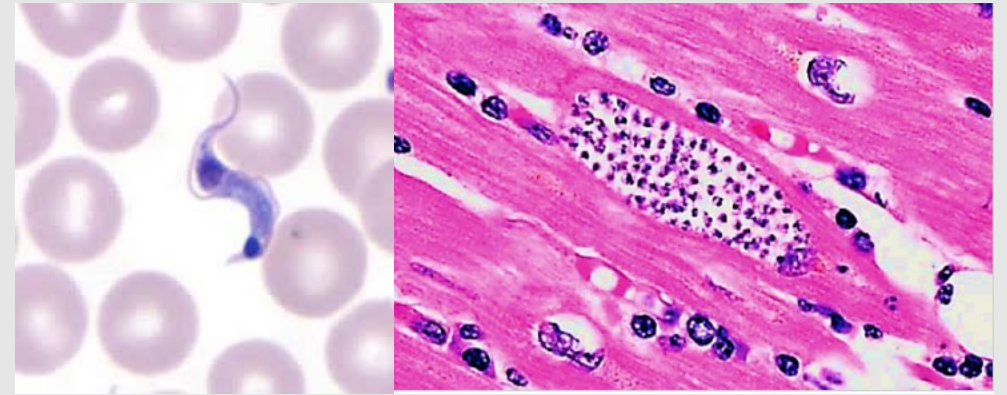
Fungal Myocarditis

- Occurs in disseminated fungal infections most commonly in **immunocompromised patients**
- Risk factors include:
 - Broad spectrum Abx use
 - Corticosteroid use
 - Invasive procedures
 - HIV infection
- **Difficult to dx:** clinical signs of myocarditis are often absent or ambiguous and blood cultures often negative
- Most common fungal causes: **Candida or Aspergillus**



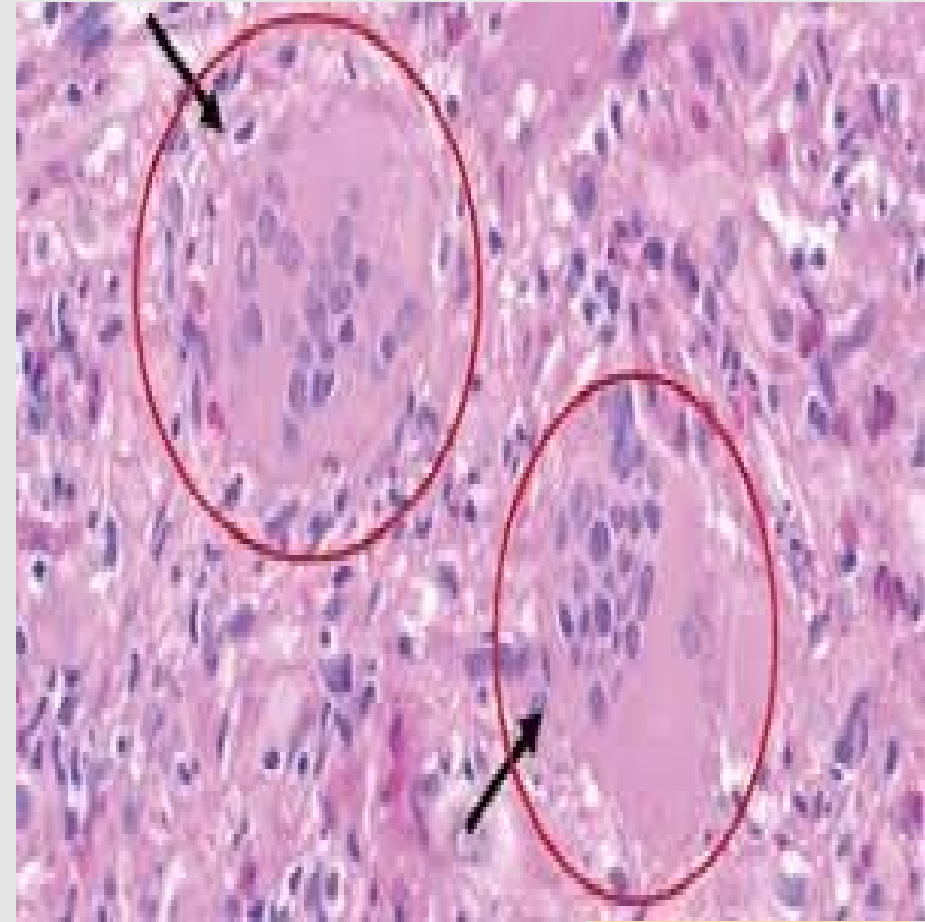
Parasitic Myocarditis

- Toxoplasma Gondii
- Trichinella
- **Trypanosoma Cruzi**
 - **Chagas Disease** – from contact with feces/urine of reduviid (kissing) bug
 - Also can be passed down from mother to fetus or from infected organ transplants/transfusions
 - Endemic in **Latin America**
 - **Pseudocyst dilations within myocardial fibers**
 - Acute Phase: lasts from 8 to 12 weeks; Chagoma (Romaña's sign) fever, edema, lymphadenopathy, anemia, hepatosplenomegaly, ECG abnormalities, CNS involvement, death
 - Indeterminate Phase: asymptomatic
 - Chronic Phase: fever, **Cardiomegaly**, ECG abnormalities, **megaesophagus**, **megacolon**



Immune Mediated Myocarditis

- **Post-Viral**
- Hypersensitivity Myocarditis
 - High proportion of eosinophils
 - Often triggered by medications
- Antibiotics, Antihypertensives, Diuretics
 - **Ampicillin, thiazides**, and mood stabilizers like lithium are specific examples that can all produce type 1 hypersensitivity reaction dominated by eosinophils
- **Giant-Cell Myocarditis:** multinucleate giant cells interspersed with lymphocytes, eosinophils, plasma cells, and macrophages; Very poor prognosis



Immune Mediated Myocarditis Cont.

- Transplant Rejection
- **Post-Streptococcal Infection (Rheumatic Fever)**
 - Molecular mimicry cross reactivity between bacterial M protein with human cardiac myosin as well as NABG with valvular tissue
 - **Aschoff body formation**
 - Granulomas associated with rheumatic fever
 - **Anitschkow cells (caterpillar cells)**
 - Macrophages within aschoff bodies that has an ovoid nucleus
- Systemic Disorders
 - SLE, Celiac disease
 - Hyper or hypothyroidism
 - IBD
 - Sarcoidosis
 - All can develop an autoimmune response to the heart through increased inflammatory cells and cytokines



Direct Cardiotoxins

- Anthracyclines
 - Ex. Doxorubicin, epirubicin, idarubicin
- Cocaine
- Alcohol
- Arsenic
- Cyclophosphamide
- Radiation
- Heavy metals





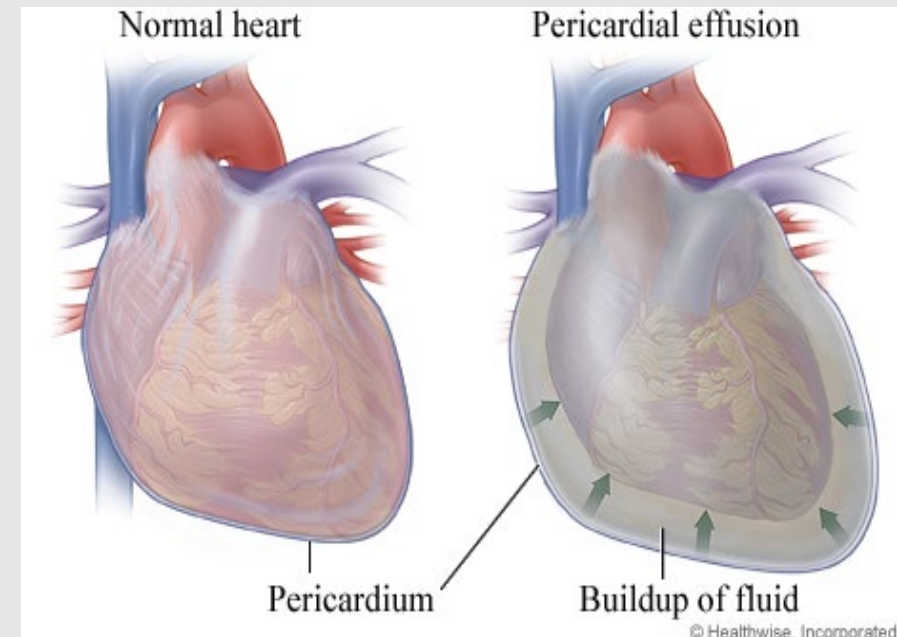
Pericardial Disease

- Involves the pericardium (the sac-like structure surrounding the heart) and/or the pericardial space
- Can occur **with direct disease of the heart or indirectly through systemic disease** (ex. Systemic lupus erythematosus)
- Involves fluid accumulation (pericardial effusion), inflammation (pericarditis), fibrous constriction or any combo of these

Pericardial Effusion

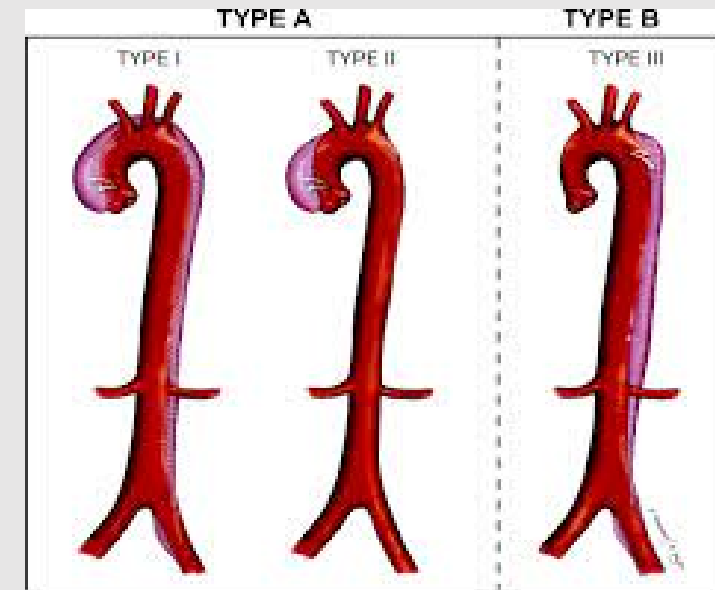


- Normal: 50 ml of straw colored (serous) fluid surrounds the heart in the pericardial space
- Disease can lead to distention of the pericardium from:
 - Excess serous fluid (pericardial effusion)
 - Accumulation of blood (hemopericardium)
 - Build up of pus (purulent pericarditis)
 - Accumulation of lymphatic fluid (chylous effusion)



Pericardial Effusion Cont.

- **Chronic/slow fluid accumulation = pericardium can remodel**
 - Allows cardiac function to be maintained
 - If chronic and <500 mL only clinical sign may be enlarged heart shadow on x-ray (water bottle configuration)
- **Acute/rapid fluid accumulation = clinically devastating**
 - With as little as 200-300 mL of fluid
 - Notably from hemopericardium (ruptured MI or aortic dissection) leading to cardiac tamponade
 - Can compress thin-walled atria/vena cava or even ventricles causing cardiovascular collapse



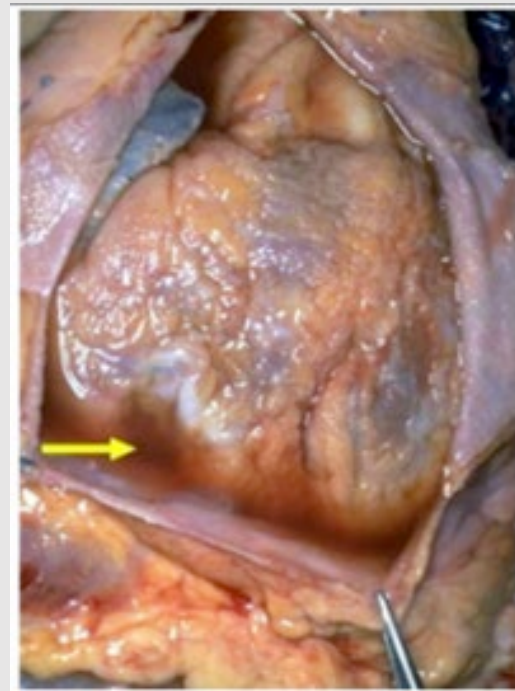


Acute Pericarditis

1. Serous Pericarditis
2. Serofibrinous Pericarditis
3. Fibrinous Pericarditis
4. Purulent/Suppurative Pericarditis

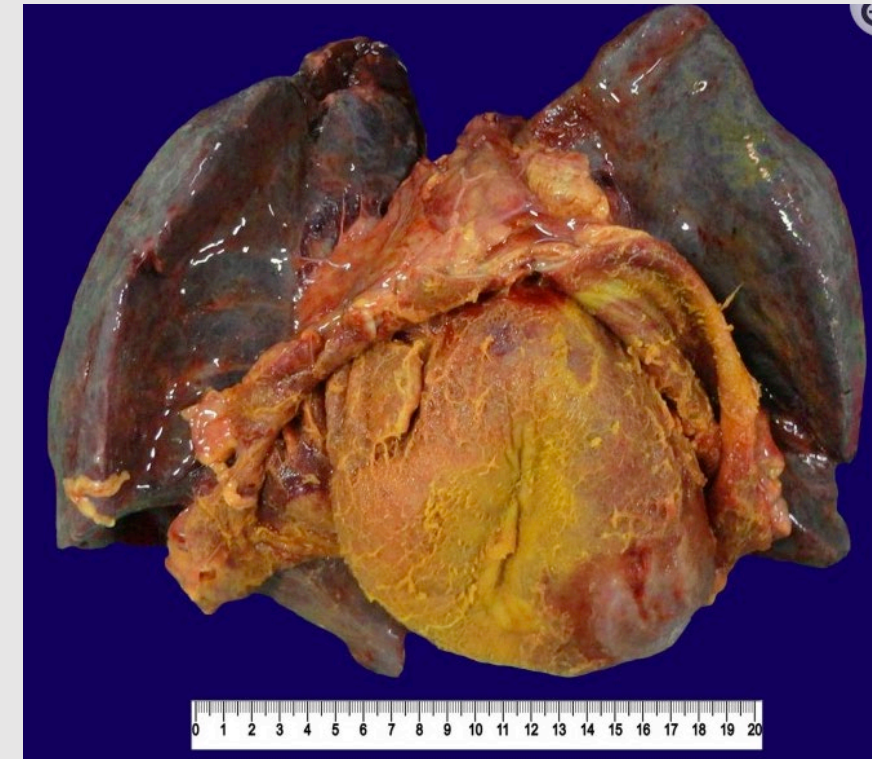
Serous Pericarditis

- Produced by:
 - **Noninfectious Inflammatory diseases** (most common)
 - Rheumatic fever, SLE, rheumatoid arthritis, scleroderma, tumors and Uremia
 - **Irritation** from infection of contiguous tissues
 - Ex. Bacterial pleuritis
 - May also progress to serofibrinous or even suppurative reactions by extension
 - Viral infections
 - Commonly person has a URI or viral infection of lung; not direct infection of pericardium
 - Can rarely directly infect pericardium; associated with viral myocarditis; termed myopericarditis
 - Tumors
 - Lymphatic invasion
 - Histology: mild inflammatory infiltrate in epipericardial fat consisting of lymphocytes; may contain neoplastic cells if associated with tumors



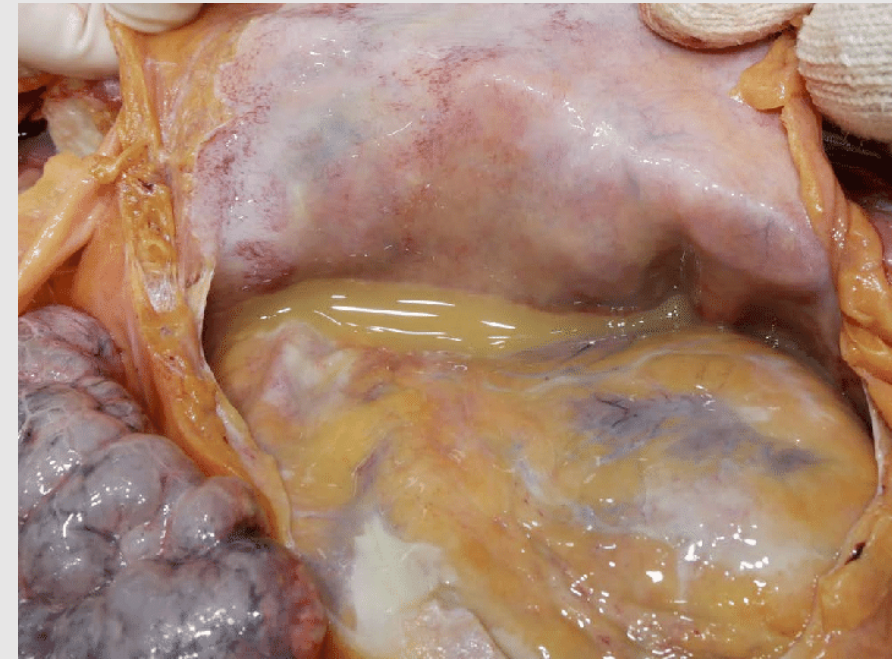
Serofibrinous/Fibrinous Pericarditis

- **Most common type** of pericarditis
- Composed of **serous fluid with fibrinous exudate**
- Macroscopically termed "**bread and butter**" due to the fibrin deposition pattern
- Causes
 - Acute MI
 - **Dressler Syndrome**
 - Immune mediated secondary pericarditis following MI, trauma, or surgery of the heart
 - Uremia
 - Rheumatic fever
 - SLE
 - Trauma
 - Can follow cardiac surgery
- Clinically: sharp chest pain that is position dependent (tripod position), **Pericardial Friction Rub (must be fibrinous)**, fever



Purulent/Suppurative Pericarditis

- **Active infection** in the pericardial space
 - Direct extension from neighboring infections
 - Pneumonia, mediastinal infections, abscesses of myocardium etc.
 - Seeding from blood
 - Lymphatic extension
 - Direct introduction from trauma/surgery



Hemorrhagic Pericarditis

- Composed of an **exudate of blood mixed with fibrinous or suppurative effusion**
- Most caused by spread of malignancy to the pericardial space; **pericardial tap often reveals neoplastic cells**
 - Also: bacterial infections, bleeding diathesis, surgery and TB





Chronic/Healed Pericarditis

- Adhesive Mediastinopericarditis
- Constrictive Pericarditis

-**Level of fibrosis ranges** from unobstrusive fibrous thickenings called soldier's plaques or thin adhesions to meshlike adhesions that largely obliterate the pericardial space

-**In most forms, cardiac function is preserved** since the pericardial sac has time to remodel.



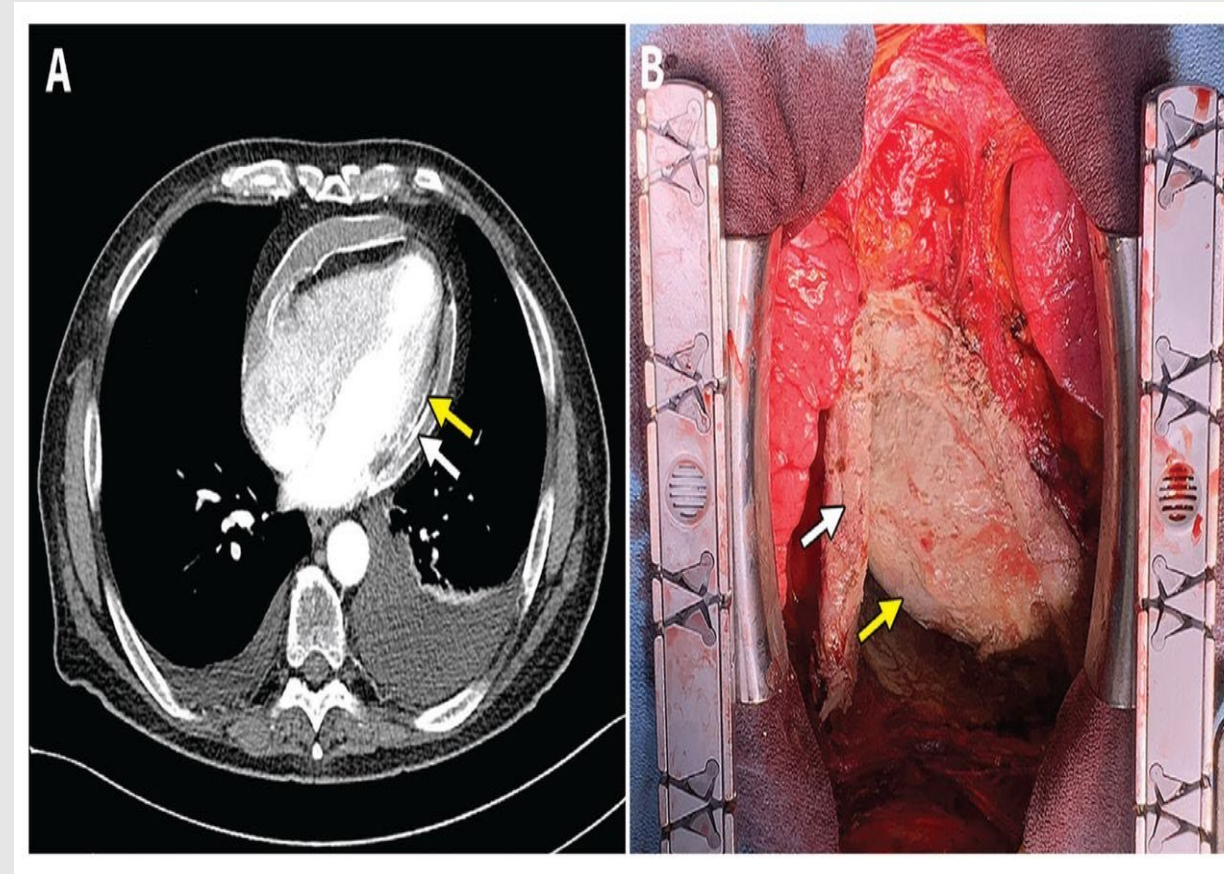
Adhesive Mediastinopericarditis

- Follows infectious pericarditis, previous surgery, or radiation
- The pericardial sac is obliterated and there are **adhesions of the external aspect of the parietal layer** that strains cardiac function
- Each heart contraction **pulls on the parietal pericardium and also attached surrounding structures**
- Clinical: systolic retraction of rib cage and diaphragm and pulsus paradoxus
 - May lead to cardiac hypertrophy and dilation from increased workload



Constrictive Pericarditis

- **Heart encased in dense, fibrous, or fibrocalcific scar** that limits diastolic expansion and cardiac output
- **Mimics restrictive cardiomyopathy**
- Scar can in some cases be 1 cm in thickness and can resemble a plaster mold (concretio cordis)
- Cardiac hypertrophy and dilation cannot occur due to the dense enclosing scar
- Heart cannot adapt to increased systemic demands



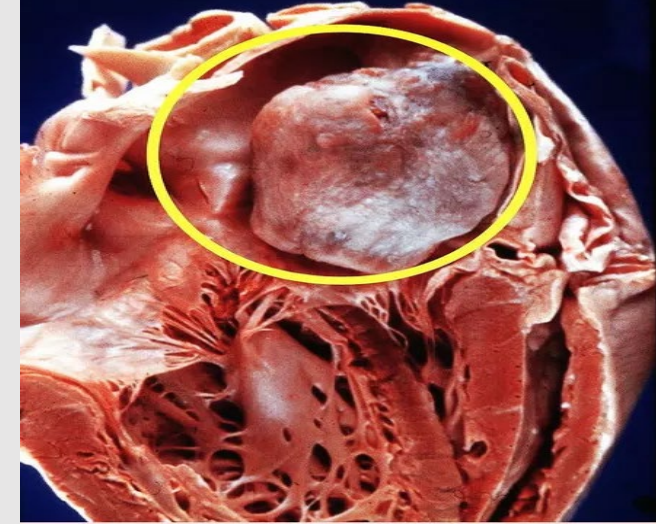


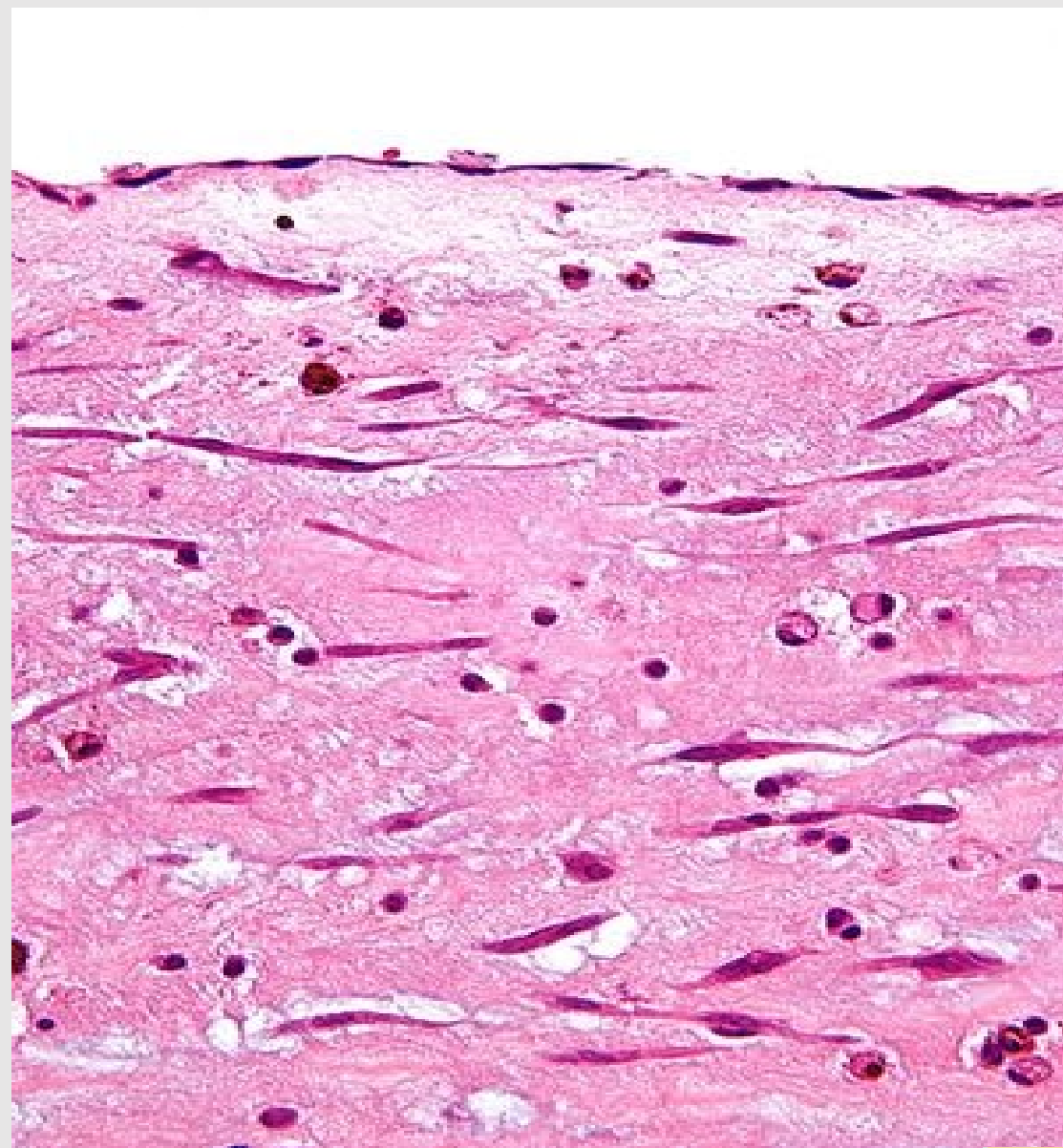
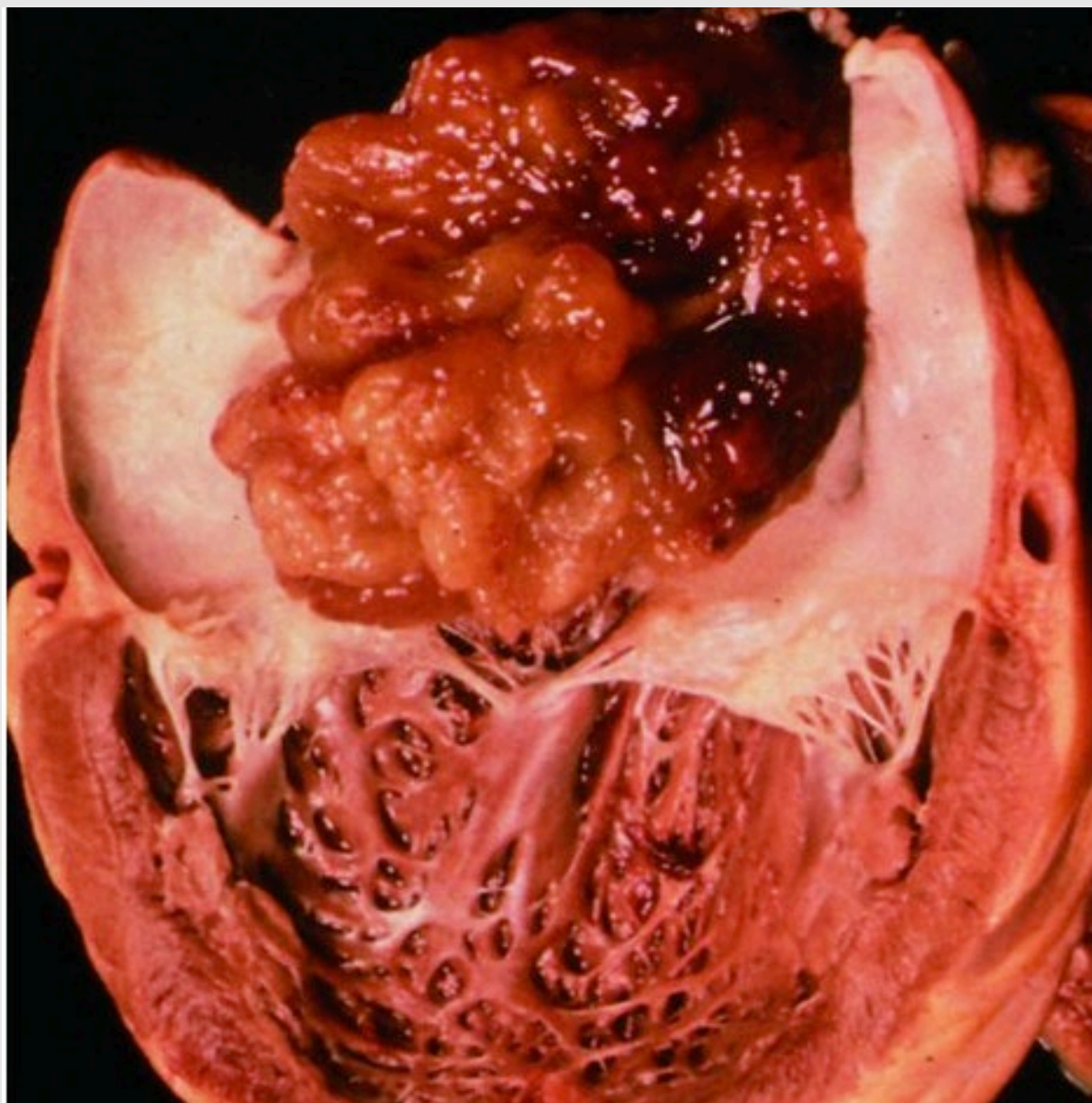
Cardiac Tumors: Overview

- Primary tumors are rare; metastatic tumors are 100x more likely
- Of the primary tumors 90% are benign
- In descending order of frequency:
 - Myxomas
 - Fibromas
 - Lipomas
 - Papillary fibroelastomas
 - Rhabdomyomas
- Angiosarcomas are the most common malignant primary cardiac tumor

Myxomas

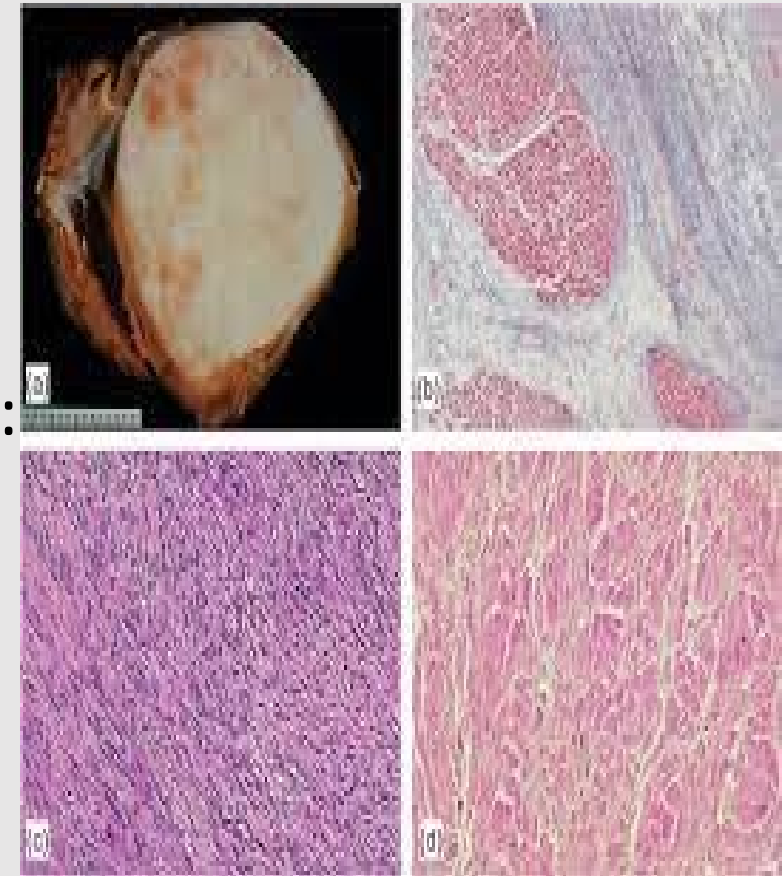
- 50-80% of primary cardiac tumors: **Most common primary cardiac tumor**
- Linked to germline mutations of PRKAR1A (alpha regulatory subunit of cAMP-dependent protein kinase type 1) which is a tumor suppressor gene (**Carney Complex**)
- Thought to arise from **multipotent mesenchymal stem cells** in the endocardium (cells of tumor have characteristics of primitive cardiac myocytes and endothelial cell precursors)
- Most (75%) occur in the **left atrium** but may occur in any chamber or on any valve.
- Clinical: obstruction, thromboembolism, conduction anomalies
- Histology: stellated fusiform and polygonal cells, immersed in amorphous myxoid matrix





Cardiac Fibroma

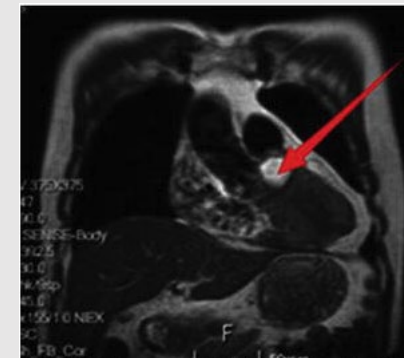
- **2nd most common primary cardiac tumor in children** behind rhabdomyoma
- Composed of connective tissue and fibroblasts, most commonly in **Left Ventricle**
- Symptoms arise due to obstruction, coronary artery compromise, thromboembolic events, and conduction defects
- Associated with **Nevoid Basal-Cell Carcinoma Syndrome**: (develop basal cell carcinomas, odontogenic keratocysts, have skeletal abnormalities)
 - Mutations in PTCH1: tumor suppressor on chromosome 9





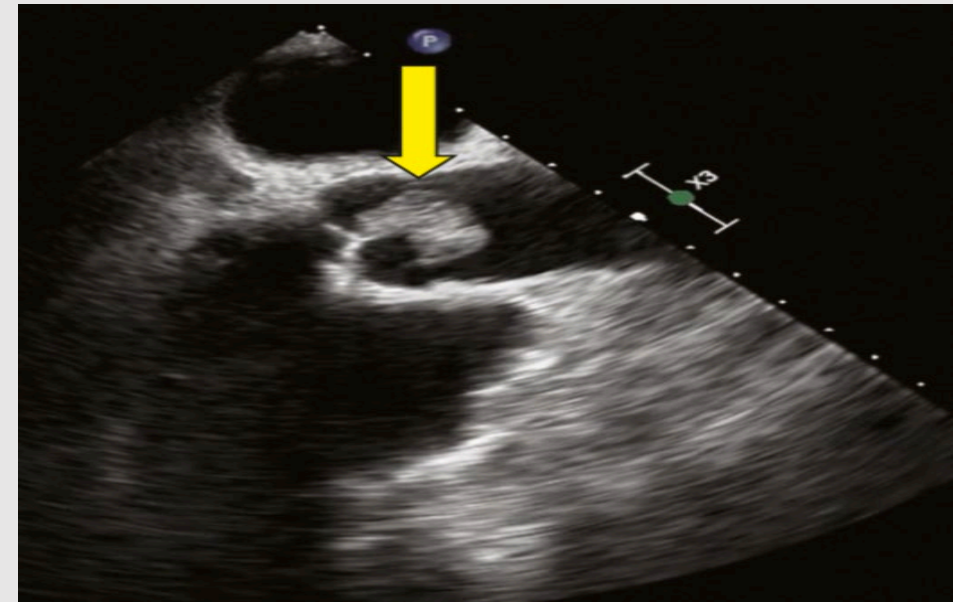
Cardiac Lipomas

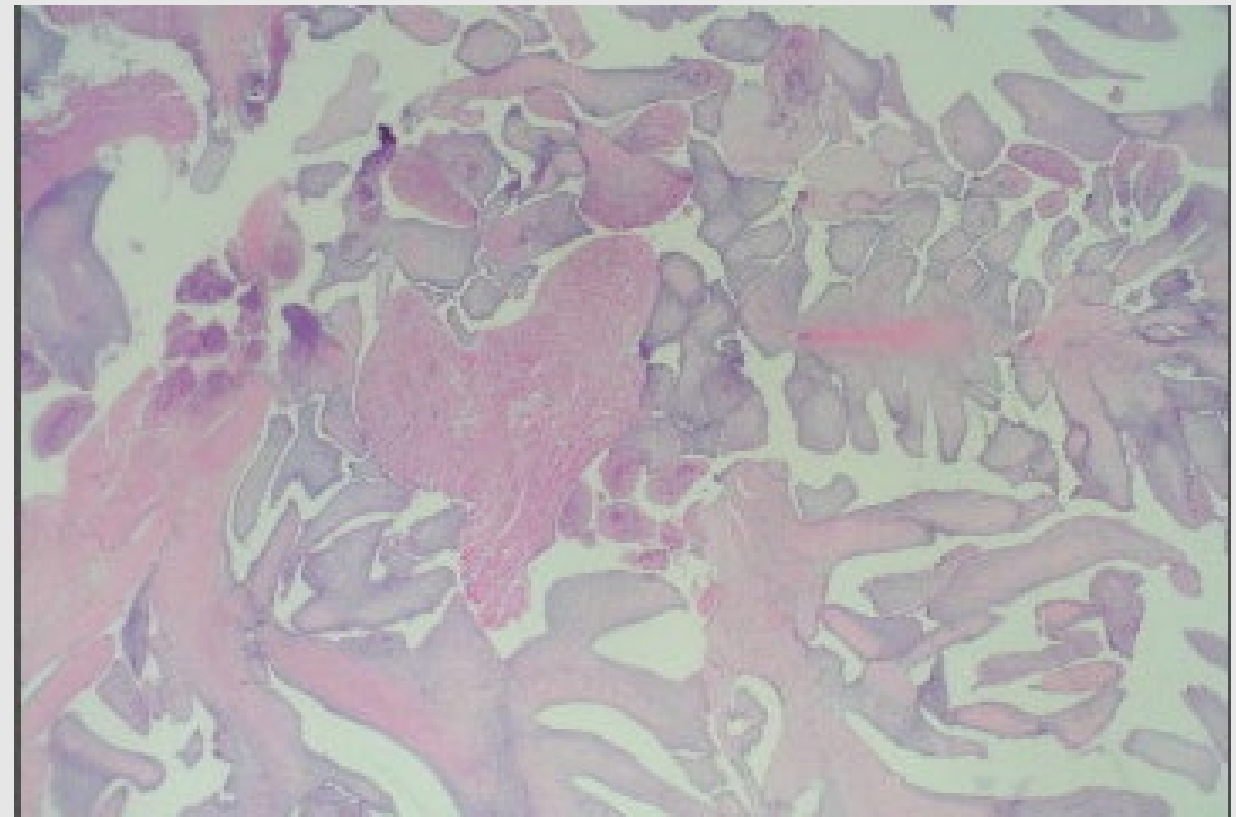
- Consists of **mature adipocytes** on histology
- Tend to occur in the **left ventricle or right atrium** but can be found anywhere in the heart and pericardium
- Frequently asymptomatic but may grow large leading to obstructive symptoms
- In rare cases can compress the SVC if it involves the right atrium



Papillary Fibroelastoma

- Considered hamartomas
- Have papillary fronds with central dense cores of collagen and elastic fibers surrounded by looser connective tissue. Surrounded by a layer of endothelial cells
- **FORM ON VALVES** measuring up to 3 to 4 cm
- Do not usually cause clinical problems but **may embolize. May also occlude coronary artery orifices** leading to myocardial ischemia
- Left sided PFs tend to be removed while the decision to remove right sided PFs is more difficult



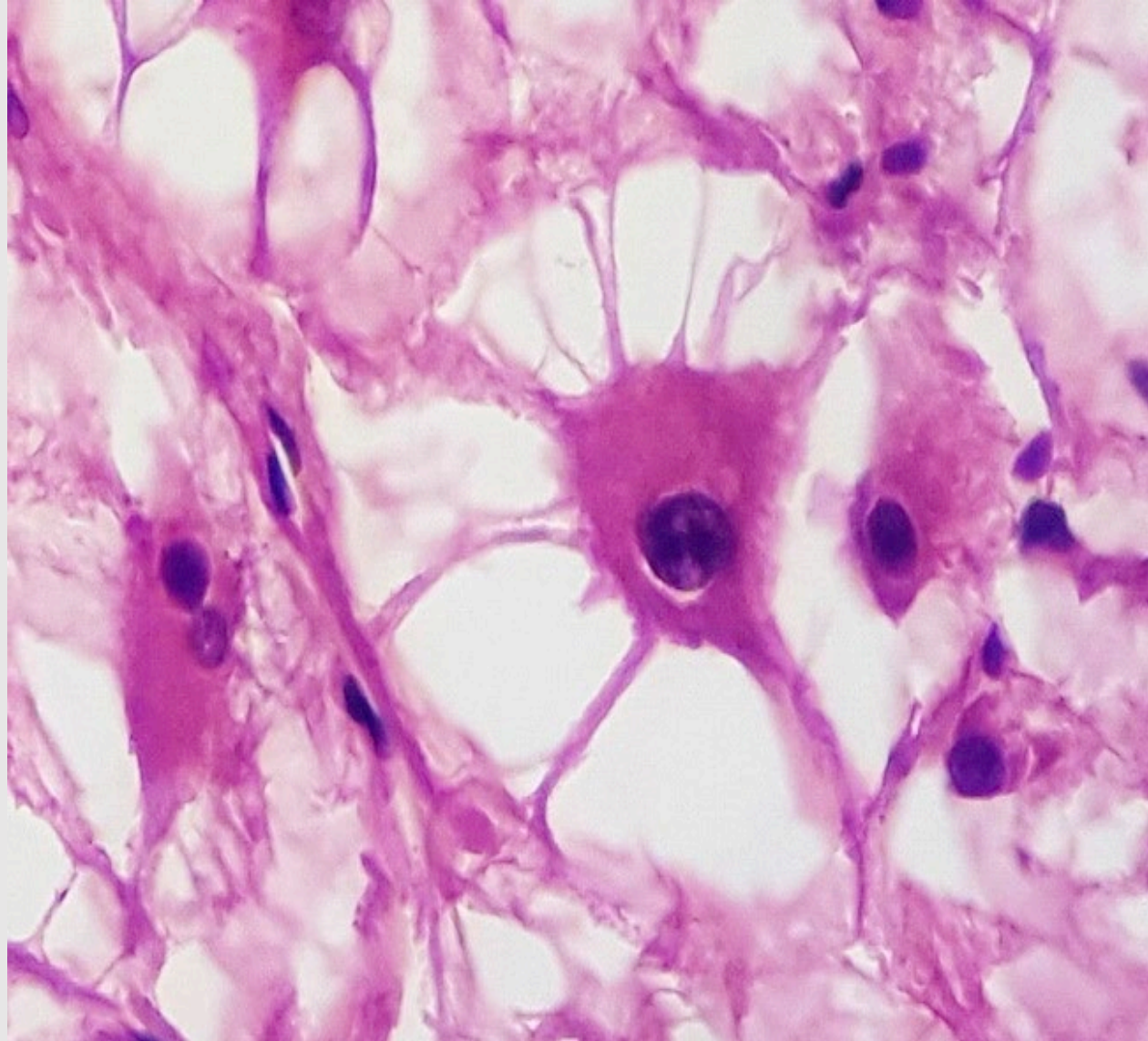


Sea anemone appearance



Cardiac Rhabdomyomas

- Most common primary **childhood** cardiac tumors
- May really be hamartomas, associated with **TUBEROUS SCLEROSIS** (mutations in TSC1 and TSC2)
- Form nodular myocardial masses, almost never found singular and involve both ventricles and in some cases the atria as well
- Histology: **SPIDER CELL** - small central nuclei and abundant glycogen rich clear cytoplasm with fibrillar processes containing sarcomeres radiate to margin of cell
- Mostly asymptomatic but may present with arrhythmias and heart failure, **many regress spontaneously** over time



Other Benign Tumors



- Cystic tumor of the AV node
 - Located at the triangle of Koch in the AV nodal region
 - Primary cardiac tumor with the highest incidence of sudden death
- Hemangioma
 - Rare primary cardiac tumor with high vascularity
- Neurofibroma
 - Neurogenic tumor with plexiform and scattered spindle cells
- Teratomas
 - Diagnosed in neonates and infants; contain endo-, meso-, and neuroectodermic germinal layers
- Leiomyoma
 - Hypothesized that leiomyomas from the uterus can metastasize to the heart through the IVC; in some cases, evidence points to the tumors being primary
- Lymphangioma
 - Contains slow flow vascular malformation with lymphatic elements

Malignant Primary Cardiac Tumors: Sarcomas

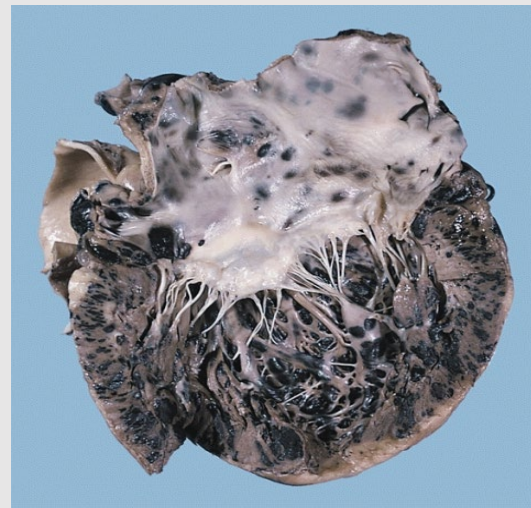
- The type of sarcoma is dependent on the tissue type
 - Ex. Angiosarcoma, rhabdomyosarcoma, leiomyosarcoma, synovial sarcoma, osteosarcoma, etc.
- **Angiosarcomas are the most common** type of cardiac sarcoma
 - They are found on the right side of the heart usually
- National Cancer Database reported that only 100,317 primary cardiac tumors from 2004 to 2016 were found of which only 826 (0.8%) were malignant and 731 were classified as sarcomas.



Metastatic Tumors



- Seen with the **most common carcinomas** such as lung, breast, GI
- Lymphomas/leukemias may also involve the heart
- Most likely cancer to metastasize to the heart is **MALIGNANT MELANOMA**
- May result in **restrictive cardiomyopathy** especially if the tumor is associated with extensive fibrosis
- **Disruptions of conduction system** may also occur leading to heart block or bundle branch blocks





Summary

- Myocarditis: range of different symptoms (asymptomatic to MI symptoms) and causes (viral, bacterial, fungal, protozoal, immune) as well as different clinical courses.
 - Dx gold standard is EMB but rarely used if H & P and other dx tests indicate myocarditis and if EMB will not alter treatment
 - May lead to dilated cardiomyopathy (restrictive if chronic active)
- Pericardial Diseases: can result from disease of the heart itself or systemic disease; may or may not be accompanied with pericardial effusion
 - Pericarditis is inflammation of the sac surrounding the heart, can be acute or chronic
- Cardiac Tumors: majority are benign in nature presenting clinically with signs of obstruction or conduction anomalies
 - Minority of primary tumors are malignant sarcomas
 - Metastases may spread to the heart and cause disruption to of the conduction system and restrictive cardiomyopathy



Lecture Feedback Form

- <https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>